

NEWINGTON COLLEGE



2014 Assessment 2 (HSC mini) Year 12 Mathematics

General Instructions:

- Date of task - Monday 31st March (Wk 10B)
- Reading time – 5 mins
- Working time – 120 mins
- Weighting - 30%
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of the paper.
- Attempt all questions.
- Show all relevant mathematical reasoning and/or calculations.

Total marks – 70

Section I (10 marks)

- Answer questions 1 to 10 on the multiple choice answer sheet provided at the end of this paper.
- Allow about 15 minutes for this section.

Section II (60 marks)

- Answer questions 11 to 14 on the writing paper provided.
- **Start each question on a new page.**
- Each page must show the candidate's computer number.

Outcomes to be assessed:

- H8** Uses techniques of integration to calculate areas and volumes.
- H3** Manipulates algebraic expressions involving logarithmic and exponential functions.
- H6&7** Uses the derivative to determine the features of the graph of a function; and uses the features of a graph to deduce information about the derivative.

Section I**10 marks****Attempt Questions 1-10****Allow about 15 minutes for this section.**

1 $\int e^{3x} dx =$

(A) $3e^{3x} + C$

(B) $e^{3x} + C$

(C) $\frac{1}{3}e^{3x} + C$

(D) $\frac{1}{3}e^x + C$

2 Convert $\frac{\pi}{7}$ radians into degrees to the nearest minute.

(A) $25^{\circ}43'$

(B) $51^{\circ}26'$

(C) $80^{\circ}47'$

(D) $0^{\circ}27'$

3 If $\log_2 x = 3$, then:

(A) $x = 9$

(B) $x = \frac{3}{2}$

(C) $x = 6$

(D) $x = 8$

4 The exact value of $\sin\left(\frac{5\pi}{4}\right)$ is:

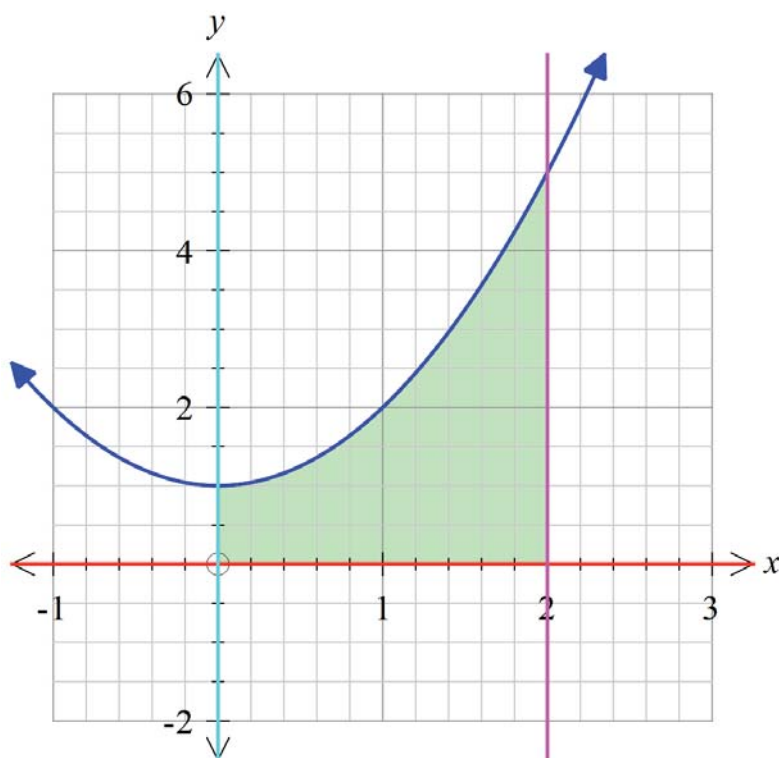
(A) $\frac{1}{\sqrt{2}}$

(B) $-\frac{1}{\sqrt{2}}$

(C) 1

(D) -1

5 Which of the following will find the shaded area on the diagram below? Note: the curve shown is $y = x^2 + 1$.



(A) $\int_0^5 2 - \sqrt{y-1} \, dy$

(B) $\int_1^5 2 - \sqrt{y-1} \, dy$

(C) $\int_0^2 x^2 + 1 \, dx$

(D) $\int_0^2 5 - (x^2 + 1) \, dx$

- 6 The gradient of the curve $y = \ln x$ at the point where $x = \frac{1}{2}$ is

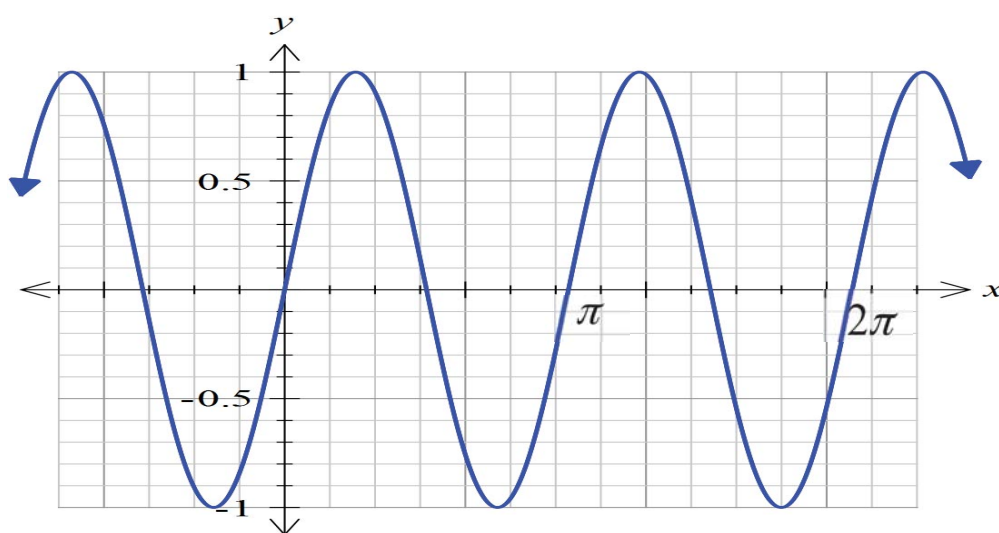
(A) $e^{\frac{1}{2}}$

(B) $\ln \frac{1}{2}$

(C) $\frac{1}{2}$

(D) 2

- 7 The graph below would be best described by which equation?



(A) $y = \sin x$

(B) $y = 2\sin x$

(C) $y = \sin 2x$

(D) $y = \sin \frac{x}{2}$

8 The derivative of $y = x^2 e^x$ is given by

(A) $\frac{dy}{dx} = 2xe^x$

(B) $\frac{dy}{dx} = x^2 e^{x-1} + 2xe^x$

(C) $\frac{dy}{dx} = 2xe^{x-1}$

(D) $\frac{dy}{dx} = x^2 e^x + 2xe^x$

9 If $2\log_a x = \log_a 9$, then

(A) $x = 3$ only

(B) $x = \pm 3$

(C) $x = 4\frac{1}{2}$

(D) $x = 81$

10 $\int_1^2 \frac{dx}{x^2} =$

(A) $\ln 4$

(B) $\frac{1}{2}$

(C) $1\frac{3}{4}$

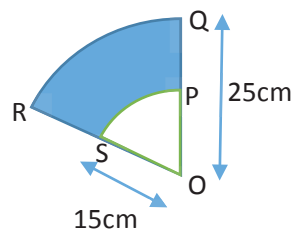
(D) $\frac{7}{24}$

End of Section I

Section II**60 marks****Attempt Questions 11-14****Allow about 1 hour and 45 minutes for this section.**

Question 11 (15 Marks) – Use a SEPARATE writing booklet.

- (a) Evaluate $\log_2 7$ to 2 decimal places. **2**
- (b) Differentiate:
- (i) $y = \frac{e^x}{x^2}$ **2**
- (ii) $y = (\ln x)^5$ **2**
- (c) Find $\int \frac{\sqrt{x+1}}{x} dx$ **2**
- (d) PS and QR are arcs of concentric circles with radii 15cm and 25cm respectively and O as the centre. The angle at the centre $\angle ROQ$ is 60° .
Diagram is not to scale.

Calculate **in terms of π** :

- (i) The area of the shaded region PQRS **3**
- (ii) The perimeter of the shaded region PQRS **2**
- (e) Sketch the curve $y = e^x + 2$ showing any asymptotes or intercepts clearly. **2**

End of Question 11

Question 12 (15 Marks) – Use a SEPARATE writing booklet.

- (a) (i) Copy and complete the table of values for $y = \sqrt{4x^2 - 1}$ in your writing booklet. **1**

x	5	5.5	6	6.5	7
y	$\sqrt{99}$	$\sqrt{120}$			

- (ii) Use Simpson's rule with 5 function values to approximate

$$\int_5^7 \sqrt{4x^2 - 1} \, dx \text{ to 3 decimal places.}$$

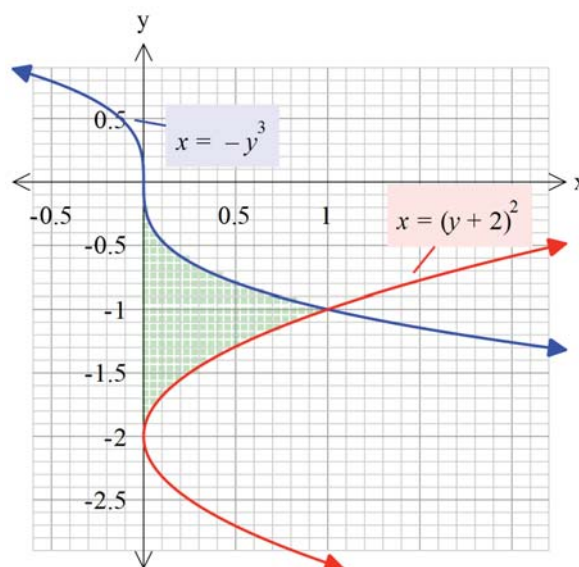
3

- (b) (i) Differentiate $e^{3x^2 - 4x}$. **1**

- (ii) Hence, find $\int (6x - 4)e^{3x^2 - 4x} \, dx$. **1**

- (c) Find the equation of the tangent to the curve $y = e^{-2x}$ at the point where $x = -1$. **4**

- (d) The diagram below shows the graphs of $x = -y^3$ and $x = (y + 2)^2$
- (i) Show algebraically that the point of intersection of the graphs $x = -y^3$ and $x = (y + 2)^2$ is $(1, -1)$. **1**
- (ii) Find the shaded area on the graph below enclosed by the curves $x = -y^3$, $x = (y + 2)^2$ and the y-axis. **4**

**End of Question 12**

Question 13 (15 Marks) – Use a SEPARATE writing booklet.

(a) Differentiate:

(i) $y = x \tan x$ **2**

(ii) $y = \ln(\sin x)$ leaving your answer in simplest form. **2**

(b) Evaluate $\int_0^{\ln 2} \frac{e^x}{1+e^x} dx$ **3**

(c) (i) Evaluate $\int_{-2}^2 x^5 dx$ **2**

(ii) Using a graph, or otherwise, explain why this integral does not give the area enclosed by this curve and the x-axis between $x = -2$ and $x = 2$. **2**

(d) Consider the function $y = 2 \cos\left(x + \frac{\pi}{2}\right)$ for $0 \leq x \leq 2\pi$.

(i) State the amplitude and the period **2**(ii) Make a neat sketch of the graph in the given domain. **2****End of Question 13**

Question 14 (15 Marks) – Use a SEPARATE writing booklet.

- (a) By writing $y = \sec x$ as $y = (\cos x)^{-1}$, show that $\frac{d}{dx}(\sec x) = \sec x \tan x$ **3**
- (b) The region enclosed by the curve $y = 2\sqrt{x}$ between $x = 1$ and $x = 4$ is rotated about the x-axis. Find the volume of the solid of revolution formed. **3**
- (c) Consider the curve $f(x) = x - \log_e x$
- (i) State the domain of $f(x)$. **1**
- (ii) Show that $f'(x) = \frac{x-1}{x}$. **2**
- (iii) By examining $f''(x)$ show that the curve is concave up for all values of x in the domain. **2**
- (iv) Find the coordinates and nature of the turning point. **2**
- (v) Sketch the curve showing clearly what happens as $x \rightarrow 0^+$. **2**

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Section 1

Solution 712 Math Mini 2014

Multiple Choice

- Q 1 (C)
- 2 (A)
- 3 (D)
- 4 (B)
- 5 (C)
- 6 (D)
- 7 (C)
- 8 (D)
- 9 (A)
- 10 (B)

Section II. Question 11

(a) $\log_2 7 = \frac{\log_e 7}{\log_e 2}$ ✓

$= \frac{2.81}{0.693} (2 \text{ dp})$ ✓

(b)(i) $\frac{dy}{dx} = \frac{x^2 e^x - 2x e^x}{x^4}$ ✓

$\left. \begin{array}{l} u = e^x \\ v = x^2 \end{array} \right\} \begin{array}{l} u' = e^x \\ v' = 2x \end{array}$ ✓

(ii) $\frac{dy}{dx} = 5(\ln x)^4 \times \frac{1}{x}$ ✓

(c) $\int \frac{x^{\frac{1}{2}+1}}{x} dx = \int x^{-\frac{1}{2}} + \frac{1}{x} dx$ ✓
 $= 2x^{\frac{1}{2}} + \ln x + c.$ ✓

(d) $60^\circ = \frac{\pi}{3}^c$ ✓

(i) $A = \frac{1}{2} r_2^2 \theta - \frac{1}{2} r_1^2 \theta$
 $= \frac{1}{2} \times \frac{\pi}{3} (25^2 - 15^2)$ ✓
 $= \frac{200\pi}{3} \text{ cm}^2$ ✓

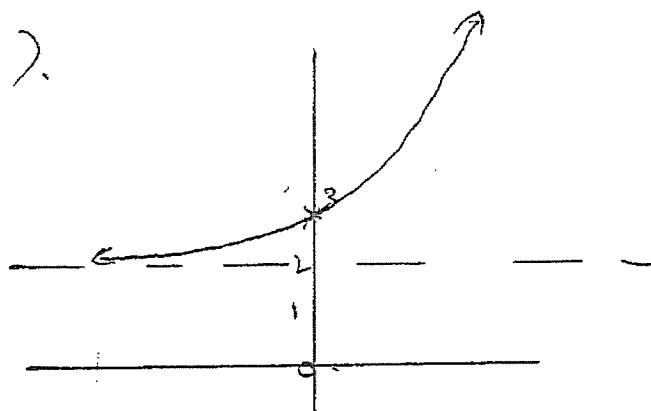
(ii) $SP = r_1 \theta$
 $= \frac{15\pi}{3}$
 $= 5\pi$

$RQ = r_2 \theta$
 $= \frac{25\pi}{3}$

$PQRS = 2 \times 10 + 5\pi + \frac{25\pi}{3}$

$= 20 + \frac{40\pi}{3}$ ✓

(e).



asymptote + intercept ✓
 shape ✓

Question 12

(a)

x	5	5.5	6	6.5	7
y	$\sqrt{99}$	$\sqrt{120}$	$\sqrt{143}$	$\sqrt{168}$	$\sqrt{195}$

$$\int_5^7 \sqrt{4x^2-1} \, dx \approx \frac{1}{3} \left\{ \sqrt{99} + \sqrt{195} + 4\{\sqrt{120} + \sqrt{168}\} + 2\sqrt{143} \right\}$$

$$= 23.957 \dots$$

$$= 23.916 \text{ (3dp)} \quad \checkmark$$

(b) (i) $\frac{d}{dx} e^{3x^2-4x} = (6x-4)e^{3x^2-4x} \quad \checkmark$

(ii) $\int (6x-4)e^{3x^2-4x} \, dx = e^{3x^2-4x} + C \quad \checkmark$

(c) $y = e^{-2x}$
 $\frac{dy}{dx} = -2e^{-2x} \quad \checkmark$

At $x = -1$ $\frac{dy}{dx} = -2e^2 \quad \checkmark$

and $y = e^2 \quad \checkmark$

Equation of tangent is $y - e^2 = -2e^2(x+1) \quad \checkmark$

(d) Sub $x = -1$ to satisfy both equations

$x = -y^3$ $x = (y+2)^3$
 LHS = 1 RAS = $-(-1)^3 = 1$ LHS = 1 RAS = $(-1+2)^3 = 1$

(ii) $\text{LHS} = \text{RAS}$ $\text{LHS} = \text{RAS}$
 $A = \int_{-1}^0 -y^3 \, dy + \int_{-2}^{-1} (y+2)^3 \, dy \quad \checkmark \checkmark$ one each integral
 $= -\left[\frac{y^4}{4} \right]_{-1}^0 + \left[\frac{(y+2)^4}{4} \right]_{-2}^{-1} \quad \checkmark$
 $= -\left(0 - \frac{1}{4}\right) + \left(\frac{1}{4} - 0\right)$
 $= \frac{1}{4} + \frac{1}{4}$
 $= \frac{2}{4} = \frac{1}{2} \quad \checkmark$

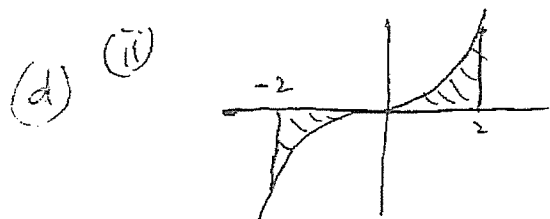
Question 13.

(a) (i) $\frac{dy}{dx} = x \sec^2 x + \tan x$ ✓✓

(ii) $\frac{dy}{dx} = \frac{1}{\sin x} \times \cos x$ ✓

(b) $\int_0^{\ln 2} \frac{e^x}{1+e^x} dx = \left[\ln(1+e^x) \right]_0^{\ln 2}$ ✓
 $= \ln(1+e^{\ln 2}) - \ln(1+e^0)$ ✓
 $= \ln 3 - \ln 2$ ✓
 $= \ln \frac{3}{2}$ ✓

(c) (i) $\int_{-2}^2 x^5 dx = 0$ ✓

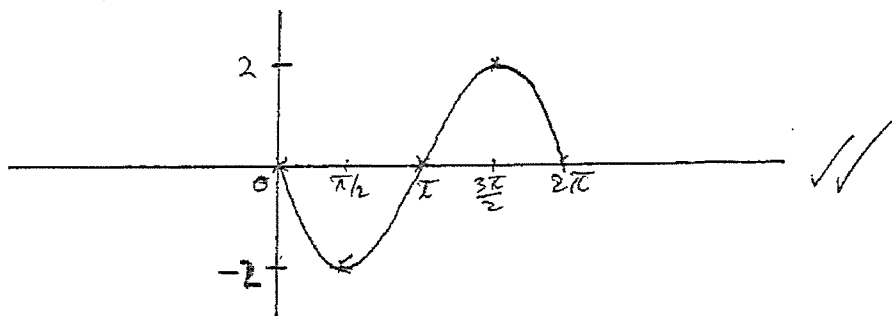


Since x^3 is odd function
shaded areas are equal.
Since area to LHS of y-axis
is below x-axis it is negative
when calculated by definite integral
Hence 'areas cancel'.

(d) (i) $y = 2 \cos(x + \frac{\pi}{2})$

amplitude = 2 ✓
period = 2π ✓

(ii) graph is cosine.. amp. 2. curve shifted $\frac{\pi}{2}$ to left.



Question 14

(a) $y = (\cos x)^{-1}$ ✓
 $\frac{dy}{dx} = -(\cos x)^{-2} \times -\sin x$ ✓
 $= + \frac{1}{\cos^2 x} \times \sin x$
 $= \frac{1}{\cos x} \times \frac{\sin x}{\cos x}$ ✓
 $= \sec x \tan x$

(b) $V = \pi \int_1^4 (2\sqrt{x})^2 dx$ ✓
 $= \pi \int_1^4 4x dx$
 $= \pi [2x^2]_1^4$ ✓
 $= \pi (32 - 2)$
 $= 30\pi$ ✓

(c) (i) $x > 0$
(ii) $f'(x) = \frac{1}{x} - \frac{1}{x}$ ✓
 $= \frac{x-1}{x}$

(iii) $f''(x) = \frac{1}{x^2} > 0$ for all x in domain ✓
since $x^2 > 0$.

∴ concave up.

(iv) turning point when $f'(x) = 0$
∴ $x = 1$ $y = 1 - \ln 1$
 $= 1$

Nature - since concave up, must be local min at $(1, 1)$

(v) As $x \rightarrow 0^+$ $y \rightarrow \infty$.

